

1. First Palindromic String in an Array and Complexity

Aim

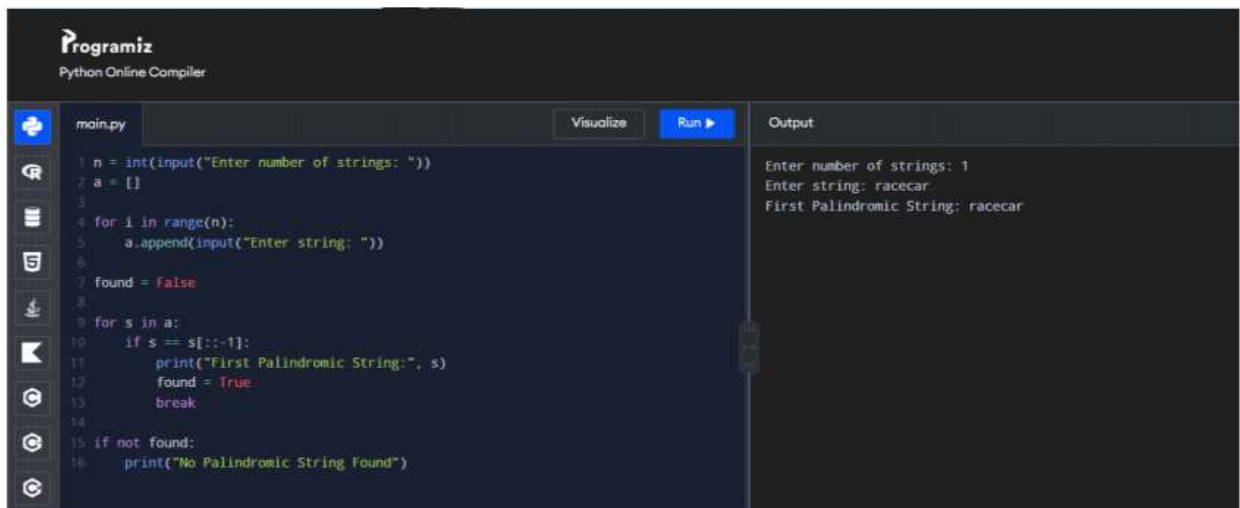
To find the **first palindromic string** in a given array of strings and analyze the time and space complexity of the algorithm.

Algorithm

1. Read an array of strings.
2. Traverse the strings one by one.
3. For each string, compare it with its reverse.
4. If both are equal, it is a palindrome.
5. Print the first palindromic string and stop.
6. If no palindrome is found, display an appropriate message.

Procedure

1. Start the program.
2. Get the number of strings and read the strings.
3. Use a loop to check each string.
4. Reverse the current string using slicing.
5. Compare the original and reversed strings.
6. If they are equal, print the string and terminate the search.
7. Display the result.



The screenshot shows the Programiz Python Online Compiler interface. The code in the editor is as follows:

```
1 n = int(input("Enter number of strings: "))
2 a = []
3
4 for i in range(n):
5     a.append(input("Enter string: "))
6
7 found = False
8
9 for s in a:
10     if s == s[::-1]:
11         print("First Palindromic String:", s)
12         found = True
13         break
14
15 if not found:
16     print("No Palindromic String Found")
```

The output on the right side of the compiler shows the following:

```
Enter number of strings: 1
Enter string: racecar
First Palindromic String: racecar
```

Result

Thus, the first palindromic string in the given array was successfully found and its time and space complexity were analyzed